

Hare-core nonemptiness for approval committees with two omissions

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Abstract

Core stability requires that no coalition can use its proportional share of seats to make every member strictly better off. We prove that every finite approval election with $m \geq 3$ candidates and $k = m - 2$ seats has an ordinary deterministic Hare-core committee, without restricting the number of voters or their approval sets. The proof bounds repeated support for blocking targets by the size of their intersection. This estimate justifies a contraction that fixes an initial blocking target, removes its strict gainers and leaves a smaller ordinary election with two omissions. Every core of the residual election lifts to a core of the original election. Induction gives a finite exact construction, including quota equality and overlapping deviations. The result concerns all candidate counts with two omissions, rather than a bounded range of committee sizes; it does not establish existence for unrestricted (m, k) .

1 Introduction

A representative committee can be challenged by voters who would all prefer to use their collective entitlement to select a different set. The Hare core rules out every such challenge when a coalition's share of seats is proportional to its share of the electorate. With binary approvals, each voter's utility is the number of approved candidates selected. The resulting stability requirement is stronger than a comparison of total approval scores: each member of a blocking coalition must strictly improve.

Proportional Approval Voting (PAV) assigns diminishing harmonic credit for each additional approved representative. Peters proved core existence for $k \leq 8$ and, using a recursive PAV procedure, for $m \leq 15$ [1, Theorems 4.1, 4.5 and 5.3]. His procedure fixes blocking targets and removes the voters who strictly gain from them before reoptimizing over full committees [1, Algorithm 1]. The result here instead fixes the number of unselected candidates: $m - k = 2$, with no upper bound on m , k or the electorate. There is no restriction on how many candidates a voter approves or disapproves.

The main step is a contraction theorem rather than a claim that PAV itself always selects a core committee. Start from a global PAV maximizer and a blocking target F . If some of its strict gainers can strictly gain again after F is selected, two omissions force their approval sets into a specific configuration. A summed swap inequality then bounds their number in terms of overlap with the later target. This bound permits a new ordinary residual election on $C \setminus F$, with fewer voters

and $k - |F|$ seats. Every residual core lifts, so no additional invariant or tie-breaking condition is required.

The proof is analytic. It does not use a finite classification, linear-programming certificate or a fractional-to-integral passage. We give the model and theorem first, establish the overlap bound, prove the contraction, and conclude by induction. An exact example then illustrates a contraction from a blocked PAV maximizer.

2 Model and main theorem

Let N be a nonempty finite voter set, $n = |N|$, and let C be a finite candidate set with $m = |C|$. Voter i approves an arbitrary set $A_i \subseteq C$. For every $X \subseteq C$, put $u_i(X) = |A_i \cap X|$. A committee has exactly k candidates, where $1 \leq k \leq m$. A nonempty coalition $S \subseteq N$ blocks a committee W by a target $T \subseteq C$ if

$$n|T| \leq k|S| \quad \text{and} \quad u_i(T) > u_i(W) \quad \text{for every } i \in S.$$

An empty target cannot strictly improve a voter, and affordability implies $|T| \leq k$. Define the full strict-gainer set and its size by

$$Q(W, T) = \{i \in N : u_i(T) > u_i(W)\}, \quad q(W, T) = |Q(W, T)|.$$

Every blocking coalition is contained in $Q(W, T)$. Conversely, if a nonempty target satisfies $kq(W, T) \geq n|T|$, its full strict-gainer set is nonempty and is itself an affordable coalition. Thus the exact core test is

$$W \text{ is core} \iff kq(W, T) < n|T| \quad \text{for every } \emptyset \neq T \subseteq C, |T| \leq k. \quad (1)$$

All quota tests below use this integer convention. Equality is blocking. No assumption of unanimous approval of a target is made, and targets may overlap the current committee.

Theorem 1. *For every finite approval election with $m \geq 3$ candidates and $k = m - 2$ seats, an ordinary deterministic Hare-core committee exists.*

Proportional Approval Voting (PAV) maximizes a harmonic approval score. We use its optimality only to initialize a contraction; a PAV maximizer need not itself be core. Define

$$H_0 = 0, \quad H_a = \sum_{j=1}^a \frac{1}{j} \quad (a \geq 1), \quad \text{PAV}(X) = \sum_{i \in N} H_{u_i(X)}.$$

A global PAV maximizer among k -committees exists by finiteness. It is also a one-swap local maximizer: replacing any selected candidate by any unselected candidate does not increase PAV.

3 Bounding repeated support

We first record an elementary removal bound. Fix a committee W , an unselected candidate b , and a nonempty set $Y \subseteq W$. For one voter, sum the PAV changes over the swaps $W - \{y\} + \{b\}$, $y \in Y$. If b is approved, every such change is nonnegative. If b is not approved and $u = u_i(W) > 0$, the sum is $-|A_i \cap Y|/u \geq -1$. If $u = 0$, the sum is zero. Thus this sum is at least -1 for every voter; no division by zero is used.

We reprove the following exclusion of Peters [1, Lemma 3.1(ii)].

Lemma 2. *A Hare blocker of a one-swap PAV local maximizer cannot introduce exactly one new candidate. In particular, if exactly two candidates are unselected, every blocker contains both of them.*

Proof. Let W be such a local maximizer and suppose $T \setminus W = \{b\}$. Write $t = |T|$ and $Y = W \setminus T$, so $d = |Y| = k - t + 1 \geq 1$. A strict gainer approves b and none of Y . Its utility at W is at most $t - 1$; hence each of the d swaps adding b and removing a member of Y increases its PAV term by at least $1/t$. Every other voter's summed contribution is at least -1 by the removal bound. Writing $q = q(W, T)$, the total change, summed over all d swaps and voters, is therefore at least

$$\frac{dq}{t} - (n - q) = \frac{(k + 1)q - nt}{t}.$$

If T blocks, $kq \geq nt$ and $q > 0$, so this expression is positive. Local maximality says each swap has nonpositive total change, a contradiction. A target contained in W has no strict gainers. When two candidates are unselected, a blocker must consequently contain both. $\square$

The next estimate holds for *every* completion of a first blocking target. Neither the later committee nor the later target needs to optimize an objective.

Lemma 3 (Repeated-support bound). *Let $|C| = k + 2$, let W_0 be a one-swap PAV local maximizer, and let F be a Hare blocker of W_0 . Put $Q_0 = Q(W_0, F)$. For any k -committee $W \supseteq F$ and any $T \subseteq C$ with $|T| \leq k$, define*

$$I = F \cap T, \quad s = |I|, \quad R = Q_0 \cap Q(W, T), \quad r = |R|.$$

Then

$$(k + 1)r \leq ns. \tag{2}$$

The later target T is not required to be affordable.

Proof. If $r = 0$, the conclusion is immediate. Suppose $r > 0$ and put $B = C \setminus W_0$, so $|B| = 2$. For $i \in R$, let $a = |A_i|$. The two strict improvements and the containment $F \subseteq W$ give

$$a \geq u_i(T) \geq u_i(W) + 1 \geq u_i(F) + 1 \geq u_i(W_0) + 2 \geq a.$$

Every inequality is therefore an equality. In particular, the voter approves both members of B , F contains exactly $a - 1$ of its approvals, and T contains all its approvals. Lemma 2 gives $B \subseteq F$, so $B \subseteq I$. The unique approved candidate d_i outside F lies in $W_0 \setminus I$, and every other approved candidate lies in I . Consequently $a \geq 3$ and $a - 1 \leq s$, in particular $s \geq 2$.

Set $Y = W_0 \setminus I$. Since both unselected candidates belong to I ,

$$d = |Y| = k - s + 2.$$

Sum the PAV changes over all $2d$ swaps adding $b \in B$ and removing $y \in Y$. For a voter $i \in R$, both additions are approved, exactly one member of Y is approved, and $u_i(W_0) = a - 2$. This voter's summed contribution is exactly

$$\frac{2(d - 1)}{a - 1} \geq \frac{2(k - s + 1)}{s}.$$

For each voter outside R , the removal bound applied separately to the two choices of b gives a summed contribution of at least -2 . Every swap has nonpositive aggregate change, so

$$0 \geq \frac{2r(k - s + 1)}{s} - 2(n - r).$$

Multiplication by the positive number $s/2$ and rearrangement yield $(k + 1)r \leq ns$, as required. $\square$

4 A core-preserving contraction

Lemma 4 (Contraction and lifting). *Let $|C| = k + 2$, let W_0 be a global PAV maximizer, and let F block W_0 . Write*

$$f = |F|, \quad Q_0 = Q(W_0, F), \quad q_0 = |Q_0|, \quad D = C \setminus F, \quad N' = N \setminus Q_0,$$

$$k' = k - f, \quad n' = n - q_0.$$

Then $k' \geq 1$, $n' \geq 1$, $|D| = k' + 2$, and $kn' \leq nk'$. Consider the ordinary residual approval election with electorate N' , candidates D , approval sets $A_i \cap D$, and k' seats. Every core committee U of this residual election lifts to an original core committee $F \cup U$.

Proof. First, $q_0 < n$: otherwise any padding of F to k candidates would strictly improve every voter over W_0 , and hence strictly increase PAV, contrary to global maximality. The original blocking inequality $kq_0 \geq nf$ then also implies $f < k$. Thus $n', k' > 0$ and

$$kn' = k(n - q_0) \leq n(k - f) = nk'. \quad (3)$$

The candidate count satisfies $|D| = k' + 2$.

Let U be a residual core committee and set $W = F \cup U$. Suppose a nonempty target T of size $t \leq k$ blocks W in the *original* election, so $kq \geq nt$ for $q = q(W, T)$. Define

$$I = F \cap T, \quad s = |I|, \quad V = T \setminus F, \quad v = |V|,$$

$$r = |Q_0 \cap Q(W, T)|, \quad q' = q - r.$$

Lemma 3 gives $(k + 1)r \leq ns$, hence $kr \leq ns$. It follows that

$$kq' = kq - kr \geq n(t - s) = nv. \quad (4)$$

The target V is nonempty, since $T \subseteq F \subseteq W$ would make strict improvement impossible. Every voter $i \in Q(W, T) \setminus Q_0$ strictly improves V over U in the residual election: cancellation of the fixed part gives

$$|A_i \cap V| > |A_i \cap U| + |A_i \cap (F \setminus T)| \geq |A_i \cap U|.$$

Combining (3) and (4),

$$kk'q' \geq nk'v \geq kn'v, \quad \text{so} \quad k'q' \geq n'v.$$

As $q' \leq n'$ and $n' > 0$, this inequality implies $v \leq k'$. Thus V is a nonempty admissible target and the q' remaining voters form an affordable strictly improving residual coalition, contradicting the core property of U . Any additional residual strict gainers only strengthen this contradiction. $\square$

The residual election is a separately defined induction instance; its quota is not substituted into an original blocking test. Every original test retains the original n, k . Equations (3)–(4) prove exactly why a hypothetical original blocker would induce a residual blocker.

5 Induction and exact construction

Proof of Theorem 1. Induct on $k \geq 1$, always with $|C| = k + 2$. For $k = 1$, choose a global PAV maximizer. Any blocker would have size one and $q = n$, hence strictly increase PAV, a contradiction.

For $k > 1$, choose a global PAV maximizer W_0 . If it is core, the claim is proved. Otherwise choose any blocking target F . Lemma 4 produces a residual ordinary approval election with a nonempty electorate, $k' + 2$ candidates and $1 \leq k' < k$ seats. The induction hypothesis gives a residual core committee U . Lemma 4 lifts it to the original core committee $F \cup U$, which has exactly $f + k' = k$ candidates. $\square$

This is also a finite construction: enumerate committees to find a global PAV maximizer; enumerate all nonempty targets of size at most the current seat count to check (1); if a blocker exists, contract as in Lemma 4, solve the smaller instance, and lift. The positive integer seat count strictly decreases at each recursive call. Any choice among tied PAV maximizers and any first blocker is allowed. No efficient running-time claim is made.

All lemmas sum over actual voters, so repeated types with positive integer multiplicities are included. Empty approvals make zero PAV changes and never strictly gain; universal approvals cannot strictly gain from a target of at most k candidates. These voters are nevertheless retained in every electorate unless they genuinely belong to a deleted full-gainer set. Finiteness is used for global maximization, exhaustive tests and the well-founded induction. There is no fractional-to-integral passage.

5.1 Contraction of a blocked PAV maximizer

Let $C = B \dot{\cup} X \dot{\cup} Y$, where $B = \{b_0, b_1\}$, $|X| = 3$ and $|Y| = 6$. There are 18 voters: five approve $X \cup \{b_0\}$, five approve $X \cup \{b_1\}$, and eight approve Y . Set $k = 9$.

The committee $W_0 = X \cup Y$ is the unique global PAV maximizer. To see this, measure the PAV loss of a committee relative to the full candidate set C . Every committee omits a pair, and its loss depends only on the pair's classes:

omitted-pair classes	BB	BX	BY	XX	XY	YY
$\text{PAV}(C) - \text{PAV}(W)$	$\frac{5}{2}$	$\frac{25}{6}$	$\frac{31}{12}$	$\frac{35}{6}$	$\frac{23}{6}$	$\frac{44}{15}$

The BB loss is strictly smallest. Nevertheless, $F = X \cup B$ blocks W_0 : the first ten voters improve from utility three to four, and

$$kq(W_0, F) = 9 \cdot 10 = 18 \cdot 5 = n|F|.$$

Under the same Hare-core and unique-global-PAV conventions, this example has fewer voters than the 27-voter example described as voter-minimal in Peters [1, Section 4.3]. We make no minimality claim for eighteen voters.

After contraction, eight voters remain, all approving the six candidates in Y , and the residual committee size is four. Every four-element subset $U \subseteq Y$ is residual core because no target of at most four candidates can strictly improve any voter. Lemma 4 therefore makes $F \cup U$ an original core committee. The accompanying code reproduces this construction and independently checks every original target, using integers for all quota tests.

6 Discussion

The two-omission condition is used at one precise point. A voter who strictly improves first to F and then to a target against a completion of F must start exactly two approvals below its total and

gain one approval at each step. This rigidity turns repeated participation into the overlap bound of Lemma 3. Counting two possible gains per voter alone would not justify deleting a coalition. The overlap bound supplies the missing charge for candidates cancelled from a later target.

The contraction preserves the class $m - k = 2$ and accepts any core of the smaller election. In each recursive call, PAV is evaluated on projected residual approval sets, with that residual election’s seat count. The lifting proof, rather than an identification of the two quotas, connects this residual calculation to every blocking test in the parent electorate. This distinguishes the argument from an assertion that an arbitrary sequence of fixed-target reoptimizations cannot overcommit seats.

With three or more omissions, two strict improvements no longer force the utility equalities used in Lemma 3. The proof therefore does not establish core existence for unrestricted (m, k) . It also makes no claim about weak Nash certificates or polynomial running time. Its conclusion is a uniform, integral existence theorem for the entire two-omission class, supported by a complete contraction-and-lifting argument.

Code availability

The accompanying package contains the exact exhaustive constructor, a separate set-based verifier and the example in Section 5.1. They use only Python’s standard library. The theorem does not depend on these diagnostic computations. Instructions and the input format are given in the package README. The repository for public release is <https://github.com/Baymax-ray/approval-core-two-omissions>.

AI use

GPT-6 Astra (OpenAI) was used for proof exploration and checking, verification-code development, manuscript drafting and revision. The author retains responsibility for the submitted content.

References

- [1] D. Peters, *The Core of Approval-Based Committee Elections with Few Seats*, Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence (IJCAI 2025), pp. 4014–4022. doi:10.24963/ijcai.2025/447.